

DAY 2: Variables, var/let/const, Scope, Hoisting, TDZ

1. Write a program to demonstrate how var, let, and const behave differently when declared inside a loop.
2. Create a program that shows the difference between function scope of var and block scope of let.
3. Write a program to prove that re-declaring a variable with var works, but not with let or const.
4. Demonstrate a case where a const object can have its properties changed, even though reassignment is not allowed.
5. Write a program showing how hoisting works differently for var vs let.
6. Create a program where accessing a let variable before declaration throws a TDZ (Temporal Dead Zone) error.
7. Demonstrate shadowing by declaring a variable inside a block that has the same name as an outer variable (with var and let).
8. Write a program showing the effect of hoisting in function declarations vs function expressions.

DAY 3: Operators

1. Write a program to check if a number is odd or even without using the modulus operator (%).
2. Use the ternary operator to classify a student's grade based on their score.
3. Demonstrate the difference between == and === by comparing variables of different data types.
4. Write a program that swaps two numbers without using a third variable, only using arithmetic operators.
5. Use logical operators to check if a year is a leap year or not.
6. Demonstrate the use of the nullish coalescing operator (??) with multiple fallback values.
7. Write a program that uses bitwise operators to check if a given number is a power of 2.
8. Write a program that calculates the area of a rectangle only if both length and breadth are greater than 0, using short-circuit evaluation (&&).

DAY 4: Conditional & Loop Statements

1. Write a program that finds the largest of three numbers using only if-else (no Math.max).
2. Use switch to build a simple calculator that can perform addition, subtraction, multiplication, and division.
3. Write a program to print all even numbers between 1 and 50 using a for loop.
4. Write a program to print the multiplication table of a given number using a while loop.
5. Use a do-while loop to repeatedly ask the user for input until they type "exit".
6. Write a program that iterates through an array of numbers and prints only the odd numbers using a for...of loop.
7. Write a program to iterate through an object and print all key-value pairs using for...in.
8. Write a program that calculates the sum of digits of a number using a loop (e.g., $123 \rightarrow 1+2+3 = 6$).

DAY 5: Functions

1. Write a named function that checks whether a string is a palindrome or not.
2. Create an arrow function that returns the factorial of a given number.
3. Write an anonymous function that sorts an array of numbers in ascending order.
4. Create a higher-order function that takes another function as input and executes it 3 times.
5. Write a program where a function returns another function that adds a specific number to its argument (closure).
6. Demonstrate first-class functions by storing multiple functions in an array and calling them one by one.
7. Write a function that takes two numbers and a callback function, and applies the callback to those numbers (e.g., addition, subtraction).
8. Create a program where you use a default parameter to calculate compound interest (default interest rate if not provided).

DAY 6: Strings & String Methods

1. Write a program to check whether a given string is a palindrome using built-in string methods.
2. Create a program that counts the number of vowels and consonants in a string.
3. Write a program that reverses a string without using loops (only string methods).
4. Use slice() and substring() to extract the middle 3 characters of a given string.
5. Write a program that replaces all spaces in a sentence with -.
6. Write a program that capitalizes the first letter of every word in a sentence.
7. Write a program to check if one string is a substring of another using includes().
8. Write a program to count how many times a specific word occurs in a paragraph using split()

DAY 7: Strings & String Methods (Continued)

1. Write a program to sort the characters of a string alphabetically.
2. Write a program that removes all duplicate characters from a string.
3. Create a program that checks whether two strings are anagrams of each other.
4. Write a program to mask the last 4 digits of a phone number (e.g., 9876543210 $\rightarrow$ *****3210).
5. Write a program that finds the longest word in a sentence.
6. Write a program that trims leading and trailing spaces from a string and counts the actual length.
7. Write a program to convert a string into title case (every word starts with uppercase).
8. Write a program that checks if a given string starts and ends with the same character.

DAY 8: Arrays & Array Methods

1. Write a program that finds the largest and smallest elements in an array.
2. Write a program that removes all duplicate elements from an array.
3. Create a program to reverse an array without using reverse() method.
4. Write a program to find the sum and average of all numbers in an array.
5. Write a program to find the index of the first occurrence of a given element in an array.
6. Write a program that joins all elements of an array into a single string separated by commas.
7. Write a program to merge two arrays and remove duplicates.
8. Write a program to find all even numbers from an array using filter().

DAY 9: Arrays & Array Methods (Continued)

1. Write a program that sorts an array of numbers in ascending and descending order.
2. Write a program to count how many times each element appears in an array.
3. Write a program to flatten a nested array (e.g., [1, [2, [3]]] → [1,2,3]).
4. Write a program to rotate an array by k positions to the right.
5. Write a program that finds the intersection of two arrays.
6. Write a program to split an array into chunks of size n.
7. Write a program that squares each element of an array using map().
8. Write a program to check if all elements in an array are unique.

DAY 10: Concept of Array-Objects

1. Write a program to create an array of objects representing students with name and marks, and print only the names of students who scored more than 50.
2. Write a program that sorts an array of objects by a specific property (e.g., age).
3. Write a program to find the student with the highest marks from an array of student objects.
4. Write a program to group an array of objects by a property (e.g., group employees by department).
5. Write a program that finds the average marks of all students in an array of objects.
6. Write a program that filters out employees earning less than a certain salary from an array of employee objects.
7. Write a program that updates the age of a person in an array of objects, given their name.
8. Write a program that checks if all products in an array of objects are in stock (inStock = true).

DAY 11: Date and Number Methods

1. Write a program to display the current date in dd-mm-yyyy format.
2. Write a program that calculates the number of days left until New Year.
3. Write a program to find the difference in days between two given dates.
4. Write a program that extracts the current hour and prints whether it's AM or PM.
5. Write a program that generates a random number between 1 and 100 using number methods.
6. Write a program that rounds a decimal number to 2 places using toFixed().
7. Write a program to check if a given number is an integer using a built-in method.
8. Write a program to convert a string "123.45" into a number and round it down using Math.floor().

DAY 12: Object, this, Call, Apply & Bind

1. Write a program to create an object car with properties and a method that prints this.brand.
2. Write a program to demonstrate how this behaves differently in a normal function vs an arrow function.
3. Create an object person with a method greet(), then borrow that method into another object using call().
4. Write a program where you pass multiple arguments to a method using apply().
5. Create a function that uses bind() to permanently bind this to a specific object.
6. Write a program where a function defined outside an object is borrowed by multiple objects using call().
7. Write a program to demonstrate losing this inside a setTimeout() function and fix it using bind().
8. Create an object method that uses this and test it when assigned to a variable (to show undefined behavior without bind).

DAY 13: Rest Parameter & Spread Operator

1. Write a function that takes an unknown number of arguments using the rest parameter and returns their sum.
 2. Write a function that finds the maximum number from an array using the spread operator.
 3. Write a program that merges two arrays using the spread operator.
 4. Write a program that copies an object using the spread operator (deep vs shallow copy case).
 5. Write a function with rest parameters that calculates the average of any number of inputs.
 6. Write a program that combines multiple objects into one using the spread operator.
 7. Write a program to remove duplicate values from an array using spread and Set.
 8. Write a program where you pass extra arguments to a function and collect unused ones using rest parameters.
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DAY 14: Array & Object Destructuring

1. Write a program that swaps two variables using array destructuring.
 2. Write a program that extracts the first two elements of an array and stores the rest in another array using destructuring.
 3. Write a program to destructure nested arrays and print individual values.
 4. Write a program that extracts specific properties (name, age) from an object using destructuring.
 5. Write a program that destructures an object and assigns default values if a property doesn't exist.
 6. Write a program to destructure a nested object (e.g., employee.address.city).
 7. Write a function that accepts an object and uses parameter destructuring to print selected fields.
 8. Write a program to swap keys and values in an object using object destructuring.
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DAY 15: JSON (stringify & parse)

1. Write a program to convert a JavaScript object into a JSON string using `JSON.stringify()`.
 2. Write a program to convert a JSON string into a JavaScript object using `JSON.parse()`.
 3. Write a program that stringifies an object with nested properties and then parses it back.
 4. Write a program that converts an array of objects into JSON and back to an array.
 5. Write a program to store a JSON string in `localStorage` and retrieve it back as an object.
 6. Write a program to filter specific properties while stringifying (using `replacer` function in `JSON.stringify()`).
 7. Write a program to pretty-print a JSON string with indentation.
 8. Write a program to fetch JSON data from a variable and print values using `JSON.parse()`.
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